

# Galois Theory and the Solution of Quintic and Sextic Polynomials over the Complex Field

An exposition for readers in algebra, group and field theory, and in computer algebra

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## Abstract

We give a self-contained account of what Galois theory determines about the solubility of quintic and sextic polynomial equations, and what modern computer algebra can and cannot compute from it. Working over the rationals with roots realized in  $\mathbb{C}$ , we recall the fundamental correspondence and the criterion that a separable equation is solvable by radicals if and only if its Galois group is a solvable group. We then carry out the classification that the criterion reduces the problem to: the five transitive subgroups of  $S_5$  and the sixteen of  $S_6$ , with their orders and solvability, each entry verified by an independent computation in GAP. We exhibit, for every transitive type we could isolate with small integer coefficients, a concrete polynomial whose Galois group we compute, together with its discriminant and the cycle types of its Frobenius conjugates at a set of primes. The computational side is developed with the precision a computer-algebra reader expects: irreducibility by modular reduction and the Lenstra-Lenstra-Lovász lattice basis reduction, factorization over finite fields, the extraction of Frobenius cycle structure from the Dedekind factorization pattern, and the use of resolvent polynomials to descend through the subgroup lattice. We close with applications — constructibility, the identification and solution of solvable quintics, and the role of Galois groups in cryptography and complexity — and a candid discussion of the limits of the method, including the subtle point that a Bring-Jerrard reduction changes the base field and hence the relevant Galois group. Every computed quantity in the paper was produced by an executable script; the accompanying log records the exact inputs and outputs.

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## 1. Introduction and the meaning of “over the complex field”

Let  $f(x) \in \mathbb{Q}[x]$  be a separable polynomial of degree  $n$  with distinct roots  $\alpha_1, \dots, \alpha_n \in \mathbb{C}$ . The phrase “solving the equation over the complex field” needs a word of caution, because it can be read in two ways, and only one of them is interesting.

Over  $\mathbb{C}$  itself, Galois theory is vacuous:  $\mathbb{C}$  is algebraically closed, so  $f$  splits completely there and the only **algebraic** extension of  $\mathbb{C}$  is  $\mathbb{C}$  itself. There is no non-trivial Galois group to study *over*  $\mathbb{C}$ . What is interesting is the Galois group over the **base field** — ordinarily  $\mathbb{Q}$  — acting on the  $n$  complex roots. Concretely, if  $L = \mathbb{Q}(\alpha_1, \dots, \alpha_n) \subseteq \mathbb{C}$  is the splitting field of  $f$  over  $\mathbb{Q}$ , then

$$G = \text{Gal}(L/\mathbb{Q})$$

is a finite group of field automorphisms of  $L$ . Each  $\sigma \in G$  permutes the roots, and the map  $\sigma \mapsto (\alpha_1, \dots, \alpha_n) \mapsto (\sigma(\alpha_1), \dots, \sigma(\alpha_n))$  embeds  $G$  faithfully and transitively into the symmetric group  $S_n$ . Thus the problem of “solving  $f$  over  $\mathbb{C}$ ” is, in the language of Galois, the problem of understanding the transitive permutation group  $G \leq S_n$  that the complex roots carry.

This reframing is the whole point of the theory. The question “can the roots be written by radicals?” — a question that looks analytic — is converted into the purely group-theoretic question “is  $G$  a solvable group?” The next section states the bridge precisely; the rest of the paper is the classification and computation that the bridge makes possible.

We take the audience to be comfortable with the definitions of field extension, splitting field, and the fundamental theorem of Galois theory, and we move quickly to the content that is specific to degrees five and six.

## 2. The bridge: solvability by radicals and solvable groups

Let  $f \in \mathbb{Q}[x]$  be separable of degree  $n$ , with splitting field  $L$  over  $\mathbb{Q}$  and Galois group  $G = \text{Gal}(L/\mathbb{Q})$ , viewed as a transitive subgroup of  $S_n$ .

**The fundamental correspondence.** For each subgroup  $H \leq G$  there is a subfield  $L^H = \{x \in L : h(x) = x \ \forall h \in H\}$ , and the assignments  $H \mapsto L^H$  and  $K \mapsto \text{Gal}(L/K)$  are inclusion-reversing bijections between the subgroups of  $G$  and the intermediate fields  $\mathbb{Q} \subseteq K \subseteq L$ . Moreover  $[L^H : \mathbb{Q}] = [G : H]$ .

**The radical criterion.** Over a field of characteristic 0:

*The roots of  $f$  can be expressed by radicals over  $\mathbb{Q}$  if and only if  $G$  is a solvable group.*

Here "solvable" means that  $G$  possesses a subnormal series  $1 = H_0 \triangleleft H_1 \triangleleft \dots \triangleleft H_r = G$  whose quotients  $H_{i+1}/H_i$  are cyclic (equivalently, abelian). The forward direction (radical expression  $\Rightarrow$  solvable  $G$ ) is the content of Abel's and Galois's work; the reverse direction (solvable  $G \Rightarrow$  radical expression) is the deeper, constructive half and is the one that makes the criterion *useful*: it says that to solve  $f$  it suffices to exhibit a tower of cyclic extensions, each of which can be unwound by the appropriate root of a norm.

For degrees  $n \leq 4$  every transitive subgroup of  $S_n$  is solvable, which is why the quadratic, cubic, and quartic formulas exist. The first degree at which a non-solvable transitive group appears is  $n = 5$ :  $A_5$  (order 60) is simple and non-abelian, so it is not solvable, and  $S_5$  (order 120) has  $A_5$  as a composition factor. This single observation is the seed of the Abel-Ruffini theorem, and it is why degrees five and six are the first where the question "is it solvable by radicals?" has a non-trivial answer.

We record the precise statement of the classical impossibility result.

**Theorem (Abel-Ruffini, Galois form).** *The general equation of degree  $n \geq 5$  is not solvable by radicals. Equivalently, the Galois group of the general polynomial  $x^n + a_{n-1}x^{n-1} + \dots + a_0$  over the rational function field  $\mathbb{Q}(a_0, \dots, a_{n-1})$  is the full symmetric group  $S_n$ , which is not solvable for  $n \geq 5$ .*

The content for the reader is not the group-theoretic fact that  $S_n$  is not solvable for  $n \geq 5$  — that is a one-line exercise once  $A_5$  is known to be simple — but the assertion that the *generic* Galois group is the full symmetric group. For  $n = 5$  this says: a "typical" quintic has Galois group  $S_5$  and is not solvable by radicals; the solvable quintics are the exceptional, structurally constrained ones.

## 3. The quintic: five transitive subgroups of $S_5$

Because  $G$  is transitive, the problem for a quintic is to identify which of the transitive subgroups of  $S_5$  it is. There are exactly five, and we give the complete list with the data that a computer-algebra system reports.

**Table 1. Transitive subgroups of  $S_5$  (verified in GAP, enumerate2.g).**

| label  | order | solvable | structure |
|--------|-------|----------|-----------|
| $5T_1$ | 5     | yes      | $C_5$     |

|        |     |           |                                                                |
|--------|-----|-----------|----------------------------------------------------------------|
| $5T_2$ | 10  | yes       | $D_{10}$<br>(dihedral)                                         |
| $5T_3$ | 20  | yes       | $C_5 : C_4(\text{Frobenius group } F_{20} = \text{AGL}(1, 5))$ |
| $5T_4$ | 60  | <b>no</b> | $A_5$                                                          |
| $5T_5$ | 120 | <b>no</b> | $S_5$                                                          |

The immediate corollary is the classical characterization, due in its computable form to Young and Dickson:

*An irreducible quintic over  $\mathbb{Q}$  is solvable by radicals if and only if its Galois group is  $C_5$ ,  $D_{10}$ , or  $F_{20}$  — that is, if and only if it is contained in the Frobenius group  $F_{20}$  of order 20.*

The three solvable cases have distinct, easily detected signatures, which is what makes them *computable*:

- $C_5$  (**cyclic**). The Galois group has order 5 and is abelian. The discriminant is a square in  $\mathbb{Q}$  (so  $G \subseteq A_5$ ) and there is no element of order 2 or 4. A cyclic quintic is not a Kummer extension of a quadratic field: the natural Kummer base for a cyclic extension of degree 5 must contain the fifth roots of unity, and that field is  $\mathbb{Q}(\zeta_5)$ , of degree 4 over  $\mathbb{Q}$  (its unique quadratic subfield is  $\mathbb{Q}(\sqrt{5})$ , which does *not* contain  $\zeta_5$ ). The correct statement is that, after adjoining  $\zeta_5$ , the compositum  $L(\zeta_5)/\mathbb{Q}(\zeta_5)$  is a Kummer extension of degree 5 (cyclic of order 5 over a field containing  $\mu_5$ ).
- $D_{10}$  (**dihedral**). Order 10,  $G \subseteq A_5$  (square discriminant), and  $G$  contains a reflection — a double transposition — but no element of order 4.
- $F_{20}$  (**Frobenius**). Order 20,  $G \not\subseteq A_5$  (non-square discriminant), and  $G$  contains an element of order 4 (a 4-cycle). This is the largest solvable transitive subgroup of  $S_5$ .

The two non-solvable cases,  $A_5$  and  $S_5$ , are separated by the discriminant:  $G \subseteq A_5$  (hence  $G = A_5$ , since  $A_5$  is the only transitive subgroup of  $A_5$  containing a 3-cycle) exactly when the discriminant is a square in  $\mathbb{Q}$ ;  $G = S_5$  when it is not. The discriminant test is the single most useful invariant for quintics, and it is cheap to compute.

**The  $A_5/S_5$  discriminator via Frobenius elements.** A further refinement, which is how the identification is actually carried out computationally, uses the Frobenius conjugates. For a prime  $p$  not dividing the discriminant of  $f$ , reduce  $f$  modulo  $p$  and factor it over  $\mathbb{F}_p$  into irreducibles of degrees  $d_1, \dots, d_k$ . The multiset  $\{d_1, \dots, d_k\}$  is the cycle structure of a Frobenius conjugate of  $G$  in  $S_5$  (Dedekind's theorem on the decomposition of primes). By sampling enough primes one collects the cycle types that occur in  $G$ , and the identification of  $G$  is then a finite matching problem against Table 1. For example, a 3-cycle appearing among the Frobenius conjugates forces  $G$  to contain a 3-cycle; combined with a square discriminant ( $G \subseteq A_5$ ) this forces  $G = A_5$ , because neither  $C_5$  nor  $D_{10}$  contains a 3-cycle.

#### 4. The sextic: sixteen transitive subgroups of $S_6$

The sextic is where the problem becomes genuinely richer. There are sixteen transitive subgroups of  $S_6$ , and the list contains phenomena that have no quintic analogue: two *distinct* embeddings of  $S_4$  into  $S_6$ , a non-solvable subgroup of order 60 that is  $A_5$  acting on six points (not on five), and the full  $S_6$  together with  $A_6$ .

**Table 2. Transitive subgroups of  $S_6$  (verified in GAP, enumerate2.g).**

| label  | order | solvable | structure |
|--------|-------|----------|-----------|
| $6T_1$ | 6     | yes      | $C_6$     |
| $6T_2$ | 6     | yes      | $S_3$     |

|           |     |           |                                                        |
|-----------|-----|-----------|--------------------------------------------------------|
| $6T_3$    | 12  | yes       | $D_{12}$                                               |
| $6T_4$    | 12  | yes       | $A_4$                                                  |
| $6T_5$    | 18  | yes       | $C_3 \times S_3$                                       |
| $6T_6$    | 24  | yes       | $C_2 \times A_4$                                       |
| $6T_7$    | 24  | yes       | $S_4$<br>(natural action on<br>$6 = \binom{4}{2}$<br>) |
| $6T_8$    | 24  | yes       | $S_4$<br>(distinct transitive<br>embedding)            |
| $6T_9$    | 36  | yes       | $S_3 \times S_3$                                       |
| $6T_{10}$ | 36  | yes       | $(C_3 \times C_3) : C_4$                               |
| $6T_{11}$ | 48  | yes       | $C_2 \times S_4$                                       |
| $6T_{12}$ | 60  | <b>no</b> | $A_5$<br>(on six points)                               |
| $6T_{13}$ | 72  | yes       | $(S_3 \times S_3) : C_2$                               |
| $6T_{14}$ | 120 | <b>no</b> | $S_5$<br>(on six points)                               |
| $6T_{15}$ | 360 | <b>no</b> | $A_6$                                                  |
| $6T_{16}$ | 720 | <b>no</b> | $S_6$                                                  |

The solvable/non-solvable split is 12 to 4. The four non-solvable groups are  $A_5$ ,  $S_5$ ,  $A_6$ , and  $S_6$  — and note that the first two act on **six** points, not on five, so a sextic with Galois group  $A_5$  is not “a quintic in disguise” but a genuinely different permutation representation.

Two features of Table 2 deserve emphasis for the computer-algebra reader:

1. **The two  $S_4$ 's.** There are two transitive subgroups of  $S_6$  isomorphic to  $S_4$ . Both have abstract group  $S_4$  and order 24, so reporting only “the group is  $S_4$ ” is an *incomplete* answer in degree 6: the abstract group does not determine the permutation representation. They are separated by the point-stabilizer structure and, equivalently, by the cycle-type set (both verified in GAP): the natural/edge action  $6T_7$  (action of  $S_4$  on the six 2-subsets of a 4-set) has point stabilizer  $C_2 \times C_2$  and cycle types  $\{(2, 2, 1, 1), (4, 2), (3, 3)\}$ ; the other representation  $6T_8$  (coset action on a cyclic subgroup of order 4) has point stabilizer  $C_4$  and cycle types  $\{(2, 2, 1, 1), (4, 1, 1, 1), (2, 2, 2), (3, 3)\}$ . The general lesson is that “computing the abstract group” is not the same as “computing the permutation representation”; in this case the cycle-type set does the separating work.
2. **The  $A_5$  and  $S_5$  on six points.** These are the transitive degree-6 representations of the two non-solvable degree-5 groups. The  $A_5$  ( $6T_{12}$ , order 60) has point stabilizer  $D_{10}$  (the dihedral group of order 10); it is the action of  $A_5$  on the cosets of its index-6 subgroup  $D_{10}$ , the normalizer of a 5-cycle. The  $S_5$  ( $6T_{14}$ , order 120) has point stabilizer  $F_{20} = C_5 : C_4$  (order 20); it is the action of  $S_5$  on the cosets of its

index-6 subgroup  $F_{20}$ , the normalizer of a 5-cycle. Both are genuinely different permutation representations from the familiar degree-5 actions, and neither is obtained from the quintic case by a simple restriction. The transitive  $S_5 \leq S_6$  is the *exotic*  $S_5$ -subgroup:  $S_6$  contains two conjugacy classes of  $S_5$ -subgroups — one intransitive (point stabilizer  $S_4$ ) and one transitive (point stabilizer  $F_{20}$ ) — and the exceptional outer automorphism of  $S_6$  interchanges them. The same phenomenon occurs for  $A_5$ : there are two conjugacy classes of  $A_5$ -subgroups of  $S_6$  (intransitive, stabilizer  $A_4$ ; transitive, stabilizer  $D_{10}$ ), again interchanged by the outer automorphism. (Note that  $\text{Out}(A_5) \cong C_2$ , not trivial:  $\text{Aut}(A_5) \cong S_5$  with  $\text{Inn}(A_5) \cong A_5$ ; the outer automorphism of  $S_6$  is a separate,  $S_6$ -specific phenomenon.) A sextic whose Galois group is this transitive  $S_5$  is not solvable by radicals, because  $S_5$  is not solvable. The point of recording both is that the sextic classification cannot be obtained by “just looking at the quintic case.”

As in the quintic, the discriminant gives the first split:  $G \subseteq A_6$  exactly when the discriminant is a square in  $\mathbb{Q}$ , which rules out  $S_5$  (on six points),  $S_6$ , and any group containing an odd permutation, and leaves  $A_5$ ,  $A_6$ , and the solvable groups. Distinguishing  $A_5$  from  $A_6$  among the square-discriminant sextics then requires cycle-type analysis. Both contain the 5-cycle  $(5, 1)$ , the double 3-cycle  $(3, 3)$ , and the double transposition  $(2, 2, 1, 1)$ , and neither contains a 6-cycle or a transposition. The separating cycle types are the 4-cycle-with-transposition  $(4, 2)$  and the single 3-cycle  $(3, 1, 1, 1)$ : the transitive  $A_5$  ( $6T_{12}$ , point stabilizer  $D_{10}$ ) contains neither, whereas  $A_6$  ( $6T_{15}$ , point stabilizer  $A_5$ ) contains both. Two directions of the argument have different logical strength. The **positive** direction is safe from a finite sample: a single observed Frobenius conjugate of type  $(4, 2)$  or  $(3, 1, 1, 1)$  forces  $G = A_6$ , because the transitive  $A_5$  contains neither. The **negative** direction is *not* safe from a finite sample: if the primes tried happen to return only cycle types in  $\{(2, 2, 1, 1), (5, 1), (3, 3)\}$ , that is consistent with  $A_6$  as well as with the transitive  $A_5$  (an  $A_6$  group can easily produce an initial sample drawn entirely from the shared cycle types), so one may not conclude  $G$  is the transitive  $A_5$  from containment of a finite observed set. That conclusion requires either a *complete* cycle-type set (guaranteed, in these two degrees, by the uniqueness of §6.1) or a resolvent certificate that the fixed field of the candidate  $A_5$  is the base field.

## 5. Verified computational examples

This section records concrete polynomials whose Galois groups we computed, with the discriminant and the Frobenius cycle types at a set of primes. Every number below is the output of an executable script (SageMath 10.9, `sagemath/ sagemath:latest`; group enumeration in GAP 4.11.1, `gapsystem/gap-docker: latest`); the inputs and the raw outputs are in the accompanying log. The Frobenius pattern at a prime  $p$  is the tuple of degrees of the irreducible factors of  $f \bmod p$ , largest first; primes dividing the discriminant are skipped because the reduction has repeated factors there.

### 5.1 Quintics — one representative per transitive type

**Table 3. Verified quintic representatives.**

| Galois group | order | polynomial                         | discriminant               | Frobenius patterns (sample primes)  |
|--------------|-------|------------------------------------|----------------------------|-------------------------------------|
| $C_5$        | 5     | $x^5 + x^4 - 4x^3 - 3x^2 + 3x + 1$ | 14641                      | $(5, )_{at 2, 3, 5, 7, 13, 17, 19}$ |
| $D_{10}$     | 10    | $x^5 - 5x + 12$                    | 64000000( <i>square</i> )  | $(5, )_{and} (2, 2, 1)$             |
| $F_{20}$     | 20    | $x^5 - 5$                          | 1953125                    | $(4, 1), (5, ), (2, 2, 1)$          |
| $A_5$        | 60    | $x^5 - 5x^4 - 5x^3 + 4x^2 + x - 5$ | 116057529( <i>square</i> ) | $(5, ), (3, 1, 1), (2, 2, 1)$       |

|       |     |                |         |                                    |
|-------|-----|----------------|---------|------------------------------------|
| $S_5$ | 120 | $x^5 - 5x - 5$ | 1153125 | $(3, 2), (5, ), (2, 2, 1), (4, 1)$ |
|-------|-----|----------------|---------|------------------------------------|

Two observations are worth making. First, the **trinomial family**  $x^5 + ax + b$  with small  $a, b$  produced representatives for  $C_5$ ,  $D_{10}$ ,  $F_{20}$ , and  $S_5$  but **no**  $A_5$  in the box searched; the  $A_5$  representative required searching general quintics, because an  $A_5$  quintic must have a square discriminant, a strong arithmetic constraint that the small trinomial box does not easily satisfy. Second, the Frobenius patterns in the right column are exactly the cycle types that the corresponding group contains: the  $C_5$  row has only 5-cycles, the  $S_5$  row has the full menu including the odd  $(3, 2)$ , and the  $A_5$  row has 3-cycles and double transpositions but no odd permutation — consistent with its square discriminant.

## 5.2 Sextics — nine of the sixteen transitive types

**Table 4. Verified sextic representatives (nine of sixteen types).**

| Galois group             | order | polynomial                         | discriminant        | Frobenius patterns<br>(sample primes)  |
|--------------------------|-------|------------------------------------|---------------------|----------------------------------------|
| $C_6$                    | 6     | $x^6 - x^3 + 1$                    | -19683              | $(6, ), (3, 3), (2, 2, 2)$             |
| $S_3$                    | 6     | $x^6 + 3$                          | -11337408           | $(2, 2, 2), (3, 3)$                    |
| $D_{12}$                 | 12    | $x^6 - 3x^3 - 1$                   | 1601613             | $(6, ), (2, 2, 2), (2, 2, 1, 1)$       |
| $C_3 \times S_3$         | 18    | $x^6 - 3x^3 + 3$                   | -177147             | $(6, ), (3, 1, 1, 1), (2, 2, 2)$       |
| $S_3 \times S_3$         | 36    | $x^6 - 3x^3 - 3$                   | 60761421            | $(6, ), (2, 2, 1, 1)$                  |
| $C_2 \times S_4$         | 48    | $x^6 - 3x + 5$                     | -143521875          | $(6, ), (4, 2), (3, 3)$                |
| $(S_3 \times S_3) : C_2$ | 72    | $x^6 - 3x + 3$                     | -9059283            | $(6, ), (3, 2, 1), (4, 2)$             |
| $A_6$                    | 360   | $x^6 - 4x^4 - x^3 - 3x^2 + 4x - 1$ | 6235009<br>(square) | $(5, 1), (4, 2), (3, 3), (2, 2, 1, 1)$ |
| $S_6$                    | 720   | $x^6 - 3x^3 - 3x - 3$              | 86383584            | $(4, 2), (3, 3), (5, 1), (3, 2, 1)$    |

The  $A_6$  row is the one that exercises the distinction made in §4: the discriminant 6235009 is a square (so  $G \subseteq A_6$ ), and the Frobenius patterns include  $(5, 1)$  and  $(3, 3)$  — the signature of  $A_6$  on six points — with no 6-cycle and no transposition anywhere in the sample. The  $S_6$  row, by contrast, has a non-square discriminant and the full range of cycle types.

The seven types we did **not** isolate with small coefficients are  $A_4$ ,  $C_2 \times A_4$ ,  $(C_3 \times C_3) : C_4$ , the two  $S_4$ 's, the  $A_5$  on six points, and the  $S_5$  on six points. Their existence and structure are recorded in Table 2 (verified in GAP); isolating a small-coefficient representative for each is a search problem that our bounded box did not complete, and we do not claim it. This is a deliberate statement of the boundary between what was *verified by direct computation* (Tables 1, 2, 3, 4) and what is asserted from the classification (the remaining entries of Table 2).

## 6. The computer-algebra view: what is actually computed

For the reader who will implement or call these routines, the following is the pipeline that the examples in §5 were produced by, with the algorithmic content and its cost stated at each step.

**Step 0 — Square-free and irreducible.** Factor out the square-free part (cheap, via the derivative). Then test irreducibility over  $\mathbb{Q}$ . The standard method reduces  $f$  modulo a suitable prime  $p$  (one not dividing the leading coefficient or the discriminant) and tests

irreducibility over  $\mathbb{F}_p$ ; if  $f$  is irreducible mod  $p$  for such a  $p$ , it is irreducible over  $\mathbb{Q}$  (the converse —  $f$  reducible over  $\mathbb{Q}$  but irreducible mod  $p$  for every  $p$  tried — is where lifting and bounds enter). Polynomial factorization over  $\mathbb{Q}$  is polynomial-time in the bit-size of the input, achieved by combining modular factorization, Hensel lifting, coefficient bounds, and lattice-reduction techniques such as the Lenstra-Lenstra-Lovász (LLL) algorithm. It is not correct to single out the LLL step as the sole reason for polynomial-time complexity; rather, LLL is the component that controls the height of the lifted factors (via a short-vector argument on the lattice of possible factor coefficients), which is what keeps the overall process polynomial in the bit-size of the input rather than exponential in the coefficient height.

**Step 1 — The discriminant.** Compute  $\Delta(f)$  and test whether it is a square in  $\mathbb{Q}$ . This single test gives the split  $G \subseteq A_n$  vs.  $G \not\subseteq A_n$ , and for a quintic it already separates the solvable groups  $(C_5, D_{10})$  from  $A_5$  (all square) and  $F_{20}, S_5$  (all non-square for the latter two, though  $F_{20}$  is the solvable non-square case). The discriminant is computed from the resultant of  $f$  and  $f'$ , or directly from the Vandermonde determinant; both are standard and polynomial-time.

**Step 2 — Frobenius cycle types.** For a batch of small primes  $p \nmid \Delta$ , factor  $f$  mod  $p$  over  $\mathbb{F}_p$  and record the multiset of degrees of the irreducible factors. Each such multiset is the cycle type of a Frobenius conjugate. Factoring over  $\mathbb{F}_p$  is polynomial-time (Cantor-Zassenhaus and its descendants). The subtlety is *how many* primes are needed: the answer depends on the gap between the cycle types of the candidate groups, and in the worst case one may need to sample until the set of observed cycle types uniquely identifies  $G$  among the transitive subgroups not yet excluded by the discriminant. In practice, for degrees 5 and 6, a dozen or so small primes suffice, as the tables above show.

**Step 3 — Resolvents and the descent.** The step that is *not* automatic from the cycle types is knowing when the cycle-type set is **complete**: a finite sample of primes returns a subset of the Frobenius conjugates, and a rare cycle type (for example the transpositions of an  $S_n$ ) may simply not have appeared yet. The **resolvent polynomials** supply the certificate that the observed set is the whole set, and more generally they drive the identification. The criterion must be stated with the correct direction and the correct acting group. Let  $G = \text{Gal}(L/K)$  act faithfully and transitively on the roots (so  $G$  is a transitive subgroup of  $S_n$ ), and let  $H \leq S_n$  be a *candidate* subgroup of the same degree (typically the point stabilizer of one of the other transitive subgroups). Choose  $\theta \in L$  whose stabilizer in  $S_n$  is exactly  $H$ , and form the resolvent

$$R(T) = \prod_{\bar{g} \in S_n/H} (T - g\theta),$$

the product over a set of distinct  $H$ -coset representatives (the distinct values  $g\theta$ ). The coefficients of  $R$  lie in  $K$ . Then

$$G \leq H^g \text{ for some } g \in S_n \iff R \text{ has a root in } K \iff R \text{ has a linear factor over } K.$$

The equivalence is the content:  $R$  has a  $K$ -root iff some conjugate  $g\theta$  lies in  $K$ , iff  $g\theta$  is fixed by all of  $G$ , iff  $G \leq \text{Stab}(g\theta) = gHg^{-1} = H^g$ . So the test answers the question “*is the actual Galois group contained in a conjugate of the candidate  $H$ ?*” — not the abstract question “ $H \leq G$ ”. This converts the subgroup-containment question into a factorization question over  $K$ , each of which is back to Step 0. The descent proceeds by stepping down the lattice of transitive subgroups — testing, at each node, whether  $G$  falls into a conjugate of the corresponding stabilizer — and it is this that makes the method complete: for fixed  $n$  the lattice is finite, so the descent terminates. In degrees 5 and 6 the cycle-type sets are unique (§6.1), so the cycle-type test is in principle already decisive; the resolvent descent is what turns a finite prime sample into a *proven* identification, and it is the engine that makes the method work for larger  $n$  as well.

**Step 4 — Group recognition.** The final identification — “this is the group  $5T_3 / 6T_{11}$ ” — is a matching of the computed invariants (order, cycle types, resolvent factorizations)

against the known list of transitive subgroups of  $S_n$ . For  $n = 5$  and  $n = 6$  these lists are Tables 1 and 2, and they are small enough to hard-code; for larger  $n$  one consults a database of transitive groups (as GAP does).

**6.1. A verified fact that simplifies degrees 5 and 6.** We computed, in GAP, the full set of cycle types occurring in each of the five transitive subgroups of  $S_5$  and each of the sixteen of  $S_6$ , and checked pairwise that **no two share a cycle-type set** (the identical-pair check returns empty in both degrees). Concretely, in degree 5 the five sets are  $\{(5)\}$ ,  $\{(5), (2, 2, 1)\}$ ,  $\{(5), (2, 2, 1), (4, 1)\}$ ,  $\{(5), (3, 1, 1), (2, 2, 1)\}$ , and  $\{(5), (3, 1, 1), (2, 2, 1), (4, 1), (3, 2)\}$  for  $C_5, D_{10}, F_{20}, A_5, S_5$  respectively — a nested chain that each step extends by exactly one new cycle type (here  $(3, 1, 1)$  is the degree-5 form of a 3-cycle; in degree 6 the same permutation type is  $(3, 1, 1, 1)$ ); in degree 6 the sixteen sets are all distinct (the table in the accompanying log lists them). The consequence is that, *for these two degrees alone*, once the set of Frobenius cycle types is known to be complete, it uniquely determines the transitive group, and the identification reduces to matching one of twenty-one fixed sets. This is a low-degree accident: it is not true in general that cycle types determine a transitive group, and the general algorithm must not rely on it. What it does mean for the computer-algebra reader is that degrees five and six are the cleanest possible setting for the Frobenius-cycle-type method, and the examples of §5 are exactly of this form.

**Complexity, stated honestly.** Several distinct complexity statements are often blurred together, and we separate them. (i) **Factorization** over  $\mathbb{Q}$  is polynomial-time in the bit-size of the input (Step 0). (ii) **Discriminant** computation is polynomial-time (Step 1). (iii) **Resolvent construction** — forming  $R(T) = \prod_{\bar{g} \in S_n/H} (T - g\theta)$  — costs polynomial-time in the degree of  $R$  and the height of its coefficients for fixed  $n$ . (iv) **Exact group determination** is the subtle step. For **fixed** degree  $n$ , there are only finitely many transitive subgroups of  $S_n$  (a constant depending on  $n$ ), each resolvent has degree bounded by a function of  $n$  alone, and each factorization is polynomial-time, so the *deterministic* resolvent-descent algorithm is polynomial in the bit-size of the coefficients. The important qualification: the *practical* method of “sampling small primes until the cycle types identify the group” (Step 2) is **not itself** a deterministic polynomial-time proof — a finite sample gives a subset of the Frobenius classes, and in the worst case an unlucky sample may miss the separating cycle type. A deterministic polynomial-time guarantee for the identification requires the resolvent descent (Step 3) to provide a certificate, not prime sampling alone. For **general**  $n$  the picture is less comfortable: the resolvent degrees grow factorially with  $n$ , the number of transitive subgroups of  $S_n$  grows rapidly, and the naive resolvent method is not polynomial in  $n$ . The complexity of computing the Galois group as a function of  $n$  is a genuine research topic, and we do not overclaim a polynomial bound in  $n$  here. What is true and useful — and what the examples in §5 instantiate — is the fixed-degree, deterministic, polynomial-time statement, which is the regime in which degrees five and six live.

## 7. The Bring-Jerrard reduction and the base-field trap

A large part of the classical theory of the quintic is organized around the **Bring-Jerrard form**  $x^5 + ax + b = 0$ . The theorem, due to Bring and Jerrard, is that *every* quintic can be transformed into Bring-Jerrard form by a Tschirnhaus transformation. The appeal is obvious: two parameters instead of five, and the form is adapted to the elliptic-function solution of Hermite.

A caution that is easy to miss, and that a computer-algebra reader should not: the statement is often (and wrongly) sharpened to “the transformation coefficients are expressible by square roots only.” That is too strong, and in the naive formulation it is false. We can see the degrees of the parameter equations directly. Write the quintic in depressed form  $t^5 + pt^3 + qt^2 + rt + s$  (the  $x^4$  term is removed by the rational shift  $x = y - p/5$ ). A **quadratic** Tschirnhaus transformation  $y = x^2 + ux + v$  kills the  $y^4$  and  $y^3$  terms; the



two conditions are linear in  $v$  and quadratic in  $u$  (after  $v = \frac{2}{5}p$  is substituted,  $u$  satisfies  $2pu^2 + 6qu + 4r - \frac{6}{5}p^2 = 0$ ), so this stage is solvable by square roots — but it leaves a  $y^2$  term, giving the form  $y^5 + Ay^2 + By + C$ , not yet  $y^5 + ay + b$ . To remove the  $y^2$  term one uses a **cubic** Tschirnhaus transformation  $z = x^3 + ux^2 + vx + w$ ; eliminating  $w$  (linear) and then  $v$  from the two remaining conditions gives, for the leading parameter  $u$ , an equation of **degree 6** — a sextic, not solvable by radicals in general. So the reduction is a *Tschirnhaus transformation*, not a radical solution of the original quintic, and the blanket “square roots only” claim conflates several distinct statements: (i) one can reduce the general quintic to Bring–Jerrard form; (ii) some preliminary Tschirnhaus steps use low-degree (radical) operations; (iii) the resulting transformation parameters have explicit algebraic constructions of bounded (but not quadratic) degree; and (iv) the reduction is *not itself* a radical solution of the general quintic. These must be kept separate.

There is, however, a second and more subtle base-field trap that a computer-algebra reader should not fall into, and we state it precisely. The Tschirnhaus transformation that produces  $a$  and  $b$  generally does **not** have rational coefficients; its parameters live over an algebraic extension of  $\mathbb{Q}$  (of the degrees just discussed, in general of degree up to 6). Consequently the Galois group of the Bring–Jerrard form  $x^5 + ax + b$  over  $\mathbb{Q}$  is not the same object as the Galois group of the original quintic over  $\mathbb{Q}$ . The clean statement is that the transformation is an isomorphism of the relevant extensions **over the field generated by the transformation parameters**, and the Galois group is preserved over that larger base field. If one computes the Galois group of  $x^5 + ax + b$  over  $\mathbb{Q}$  without remembering that  $a, b$  are not rational, one is computing the wrong group. This is not a cosmetic point: it is exactly the reason that the Bring–Jerrard reduction, though it simplifies the *equation*, does not by itself simplify the *Galois-group computation* over  $\mathbb{Q}$ . The reduction is the right tool for the *solution* (Hermite’s elliptic-function method works on the reduced form), and the wrong tool if one conflates it with the *group-theoretic classification* over the original base field.

We flag the same caution for the icosahedral method of §8: it is a solution method that changes the base field and the function field in which the inversion is performed, and its “solution” is by elliptic (hence transcendental) means, not by radicals.

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## 8. Solving the solvable cases: Hermite, Klein, and the icosahedron

For the three solvable transitive types, the radical criterion of §2 is not merely a decision procedure; it is a *construction*. The classical constructions, which we summarize because they are the historical and computational heart of the subject:

- $C_5, D_{10}, F_{20}$ . These are the groups for which the roots are expressible by radicals in the strict sense. The construction proceeds by the tower of cyclic extensions guaranteed by the criterion. In the  $F_{20}$  case the group has a normal  $C_5$  subgroup with cyclic  $C_4$  quotient, so the corresponding tower consists of a cyclic quartic subextension (itself a tower of two quadratic extensions) topped by a cyclic quintic extension; as with the  $C_5$  case, the quintic top becomes a Kummer extension once  $\zeta_5$  is adjoined. In practice, and in the computer-algebra systems, the roots of a solvable quintic are expressed by a finite combination of radicals and the roots of lower-degree resolvents; the systems (Sage, Magma, PARI) implement this via the same resolvent machinery of §6, specialized to the solvable case.
- $A_5$  (**and the general quintic**) by elliptic functions — **Hermite, 1858**. Hermite showed that the general quintic can be solved using elliptic functions: one reduces to the Bring–Jerrard form, constructs an associated elliptic curve whose  $j$ -invariant is a rational function of the coefficients, and inverts a certain elliptic modular function. The inversion is the non-algebraic step, and it is what makes this a *solution* in the broad sense (the roots are expressed in terms of known transcendental functions) rather than a radical solution.

- **The icosahedral equation — Klein, 1884.** Klein's geometric reformulation identifies the  $A_5$  that governs the general quintic with the rotation group of the icosahedron. The "icosahedral equation" is a certain equation whose Galois group is  $A_5$ , and Klein shows it can be solved by reducing to a binary icosahedral invariant problem and then inverting a modular function. The modern reading, which is the one that matters for computation, is that the icosahedral method is an *explicit* solution of the  $A_5$  case in terms of the hypergeometric and modular functions, and that the transcendental inversion can be replaced by a purely iterative (hence computable) procedure. This is the bridge between the 19th-century theory and the modern "solve by iteration" algorithms: the same  $A_5$  that Abel–Ruffini declares unsolvable *by radicals* is exactly the group that the icosahedral/elliptic method solves *by transcendental functions*.

The two threads — "unsolvable by radicals" (Abel–Ruffini) and "solvable by elliptic functions" (Hermite–Klein) — are not a contradiction; they answer different questions. The first asks whether the roots lie in a radical extension of  $\mathbb{Q}$ ; the second asks whether they can be expressed in terms of a chosen family of transcendental functions. The Galois group  $A_5$  is the invariant that both questions are about.

## 9. Applications

We sketch the principal applications, each of which is a direct consequence of the classification and the radical criterion.

**Constructibility and the classical problems.** The constructible numbers are precisely those lying in a tower of quadratic extensions of  $\mathbb{Q}$ , so the constructible degrees are powers of 2. Galois theory explains, in one line, why angle trisection (degree 3) and doubling the cube (degree 3, and in fact the relevant cubic is irreducible with a non-power-of-2 degree) are impossible with straightedge and compass: the required extensions are not towers of quadratics. The same machinery shows which regular  $n$ -gons are constructible. By the Gauss–Wantzel theorem, a regular  $n$ -gon is constructible if and only if

$$n = 2^k p_1 p_2 \cdots p_r,$$

where  $k \geq 0$  and the  $p_i$  are **distinct** Fermat primes (primes of the form  $2^{2^m} + 1$ ). The full criterion needs both the power-of-two factor and the distinctness: the cyclotomic Galois group  $(\mathbb{Z}/n\mathbb{Z})^\times$  must be a 2-group (a 2-power order, which is what straightedge-and-compass construction requires), and that happens precisely when the odd part of  $n$  is a product of distinct Fermat primes. (The currently known Fermat primes are 3, 5, 17, 257, 65537. So, for instance, the 15-gon ( $15 = 3 \cdot 5$ ) and the 17-gon are constructible, while the 9-gon ( $9 = 3^2$ , a repeated prime) and the 27-gon are not.)

**Identification and solution of solvable quintics.** Given a concrete quintic, the pipeline of §6 decides in polynomial time (in the coefficient bit-size) whether it is solvable by radicals, and if so produces the radical expression. This is a routine computation in the major systems, and it is the practical payload of the entire theory: most "solvable quintics" that appear in applications (characteristic polynomials of matrices with special structure, minimal polynomials in number fields) are detected and solved this way.

**Cryptography (a modest connection).** There is a genuine, but often overstated, connection between Galois theory and modern cryptography. The relevant object in isogeny-based and pairing-based cryptography is not the solvability-by-radicals of a low-degree polynomial, but the **Galois representations** and **Frobenius conjugacy classes** that arise in the arithmetic of elliptic curves and the algebraic structures underlying several cryptographic constructions. Concretely, the extension of a finite field generated by the torsion points of an elliptic curve (or by the field of definition of an isogeny) carries a Galois action, and the Frobenius element (or its conjugacy class) in that Galois group encodes arithmetic information used in the design and analysis of isogeny-based and pairing-based protocols. The connection is principally through representations and field extensions —

and through the structure of the relevant Galois groups — rather than through the radical solvability of quintic or sextic equations. The lesson of the quintic and sextic cases transfers in a limited but real way: the relevant group is a concrete permutation (or linear) group, and identifying it is a finite, computable problem once the right invariants are chosen. We record the connection as an application of the *machinery* (Galois groups, Frobenius classes, field extensions) rather than as a claim that solvability-by-radicals is a hardness parameter for these cryptographic problems.

**Complexity and invariants.** The Galois group is a canonical invariant of a polynomial, and the question of the complexity of computing it (as a function of the degree  $n$ ) is a standing problem at the interface of algebra and complexity theory. The fixed-degree polynomial-time result of §6 is the baseline; the general-degree problem is where the open questions live, and it is one of the places where the computer-algebra and the pure-mathematics viewpoints meet most directly.

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## 10. Limitations, caveats, and what is open

We close by separating, explicitly, what this exposition has *established* from what it has merely *described*, and by naming the limitations.

1. **The classification is complete; the small-coefficient representatives are not.** Tables 1 and 2 are complete and were verified by independent group enumeration in GAP. Tables 3 and 4 give verified representatives for all five quintic types and for nine of the sixteen sextic types. The seven sextic types without a small-coefficient representative in our search are recorded as such; we do not claim to have exhibited them by direct computation.
2. **The Bring-Jerrard and icosahedral methods change the base field.** As argued in §7, the reduced form and the icosahedral equation live over a larger base field than  $\mathbb{Q}$ , and the Galois group over that larger field is not the Galois group over  $\mathbb{Q}$  of the original. Any statement that “the quintic is solvable” by these methods must be read as a statement about the extension of the base field that the method introduces, and about a transcendental (not radical) expression.
3. **The radical criterion is an equivalence over characteristic 0.** The forward direction (radical  $\Rightarrow$  solvable group) holds in all characteristics; the reverse direction (solvable group  $\Rightarrow$  radical expression) uses the existence of the roots of unity and is the statement that is specific to characteristic 0 (or to a base field containing the relevant roots of unity). We work over  $\mathbb{Q}$ , so the equivalence is the one we use, but the caveat is worth stating for the general reader.
4. **Computing the Galois group for general degree is not solved in the polynomial-in- $n$  sense.** The fixed-degree result is clean; the general-degree complexity is open, and the naive resolvent method is factorial in the degree. We do not claim a polynomial-in- $n$  bound.
5. **The two  $S_4$ 's in degree 6 are a warning.** They have the same abstract structure ( $S_4$ ) and the same order (24), so an implementation that stops at “the group is  $S_4$ ” has computed an *incomplete* answer in degree 6. The two representations are separated by the point-stabilizer structure ( $C_2 \times C_2$  vs  $C_4$ ) and, equivalently, by the cycle-type set (§4) — but only because in degree 6 the cycle-type sets happen to be unique; the abstract group alone never suffices, and in higher degrees distinct transitive groups can share cycle-type sets, where the point stabilizers or a resolvent are then required.

The open problems, in short, are: the general-degree complexity of Galois-group computation; the explicit, uniformly effective construction of the radical expressions for the solvable cases (as opposed to their existence); and the refinement of the invariants that separate the genuinely distinct permutation representations that share an abstract group (the two  $S_4$ 's being the minimal example).

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## 11. Conclusion

Galois theory converts the analytic question “can the roots of this quintic or sextic be written by radicals?” into the algebraic question “is the transitive permutation group on the complex roots a solvable group?” For degree five the answer space is the five transitive subgroups of  $S_5$ , of which three are solvable; for degree six it is the sixteen of  $S_6$ , of which twelve are solvable. The classification is complete and was verified here by independent group enumeration, and we exhibited, by direct computation, a representative polynomial for every quintic type and for nine of the sextic types, with discriminants and Frobenius cycle types. The computer-algebra content — irreducibility by modular reduction and LLL, the discriminant test, Frobenius cycle types from the Dedekind factorization pattern, and the resolvent descent through the subgroup lattice — is polynomial-time for fixed degree, and it is the machinery that makes the classification a computation rather than merely a theorem. The limits are equally clear: the Bring–Jerrard and icosahedral solution methods change the base field and use transcendental rather than radical expressions; the small-coefficient representatives do not cover all sixteen sextic types; and the general-degree complexity of the problem remains open. Degrees five and six are, in the end, the first degrees at which a nonsolvable transitive Galois group occurs — degree four and below admit only solvable transitive groups, which is why the classical formulas stop there — and they remain the cleanest place to see the bridge between the Galois group of the complex roots and the possibility of solving the equation.

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## Appendix A. Reproducing the computations (and a corrected API note)

Every number in the paper was produced by an executable script; this appendix records the working API and the exact commands. The environment is SageMath 10.9 (sagemath/sagemath:latest) for the polynomial-side computations (Galois groups of specific polynomials, discriminants, and Frobenius cycle types from the Dedekind factorization pattern) and GAP 4.11.1 (gapsystem/gap-docker:latest) for the transitive-group enumeration.

### A.1. Native Sage — the working API (SageMath 10.9).

```
R.<x> = QQ[]                # rational polynomial ring
f = x^5 - 5*x - 5          # an example quintic
G = f.galois_group()        # returns a TransitiveGroup (or (G, gens))
if isinstance(G, tuple): G = G[0]
G.order()                  # -> 120
G.is_solvable()            # -> False
G.structure_description()   # -> "S5"
f.discriminant()           # -> 1153125
# Frobenius cycle type at a prime p not dividing the discriminant:
fs = f.change_ring(GF(p)).factor()
tuple(sorted([P.degree() for P, _ in fs], reverse=True)) # e.g. (3,2)
```

**A.1a. The Sage API is version- and object-specific (a caution).** A number of guides (and the one attached to this work) suggest `G.transitive_label()` to read off the label of the transitive group. Whether that call works depends on **which Sage object is returned** and on the **Sage version** — and this is the kind of detail that is easy to get wrong, because the *name* of a plausible method is easy to invent. In the tested SageMath 10.9 environment we probed both entry points directly:

- `Polynomial.galois_group()` returns a `TransitiveGroup` object that has `order()`, `is_solvable()`, `structure_description()`, and `transitive_number()`, but **not** `transitive_label()` (and not `cycle_type()`).

- `NumberField(...).galois_group()` returns a `GaloisGroup_v2` object that has all of the above **and** `transitive_label()` (it returned, e.g., "5T5" for  $x^5 - 5x - 5$ ).

So the "cheat sheet" is correct for the number-field API and incorrect for the polynomial API; the defensible statement is that the exact method set depends on the object returned and the Sage version, and the only reliable check is to probe the object. For the transitive label one may therefore use `NumberField(f).galois_group().transitive_label()`, or match `structure_description()` and `order()` against the verified Tables 1 and 2, or query GAP directly.

**A.1b. Exact transitive labels (verified, SageMath 10.9, number-field API).** Using `NumberField(f).galois_group().transitive_label()` we confirmed the labels of every representative in §5:

| representative                     | label     | structure_description    | order | solvable |
|------------------------------------|-----------|--------------------------|-------|----------|
| $x^5 + x^4 - 4x^3 - 3x^2 + 3x + 1$ | $5T_1$    | $C_5$                    | 5     | yes      |
| $x^5 - 5x + 12$                    | $5T_2$    | $D_5$                    | 10    | yes      |
| $x^5 - 5$                          | $5T_3$    | $C_5 : C_4$              | 20    | yes      |
| $x^5 - 5x^4 - 5x^3 + 4x^2 + x - 5$ | $5T_4$    | $A_5$                    | 60    | no       |
| $x^5 - 5x - 5$                     | $5T_5$    | $S_5$                    | 120   | no       |
| $x^6 - x^3 + 1$                    | $6T_1$    | $C_6$                    | 6     | yes      |
| $x^6 + 3$                          | $6T_2$    | $S_3$                    | 6     | yes      |
| $x^6 - 3x^3 - 1$                   | $6T_3$    | $D_6$                    | 12    | yes      |
| $x^6 - 3x^3 + 3$                   | $6T_5$    | $C_3 \times S_3$         | 18    | yes      |
| $x^6 - 3x^3 - 3$                   | $6T_9$    | $S_3 \times S_3$         | 36    | yes      |
| $x^6 - 3x + 5$                     | $6T_{11}$ | $C_2 \times S_4$         | 48    | yes      |
| $x^6 - 3x + 3$                     | $6T_{13}$ | $(S_3 \times S_3) : C_2$ | 72    | yes      |
| $x^6 - 4x^4 - x^3 - 3x^2 + 4x - 1$ | $6T_{15}$ | $A_6$                    | 360   | no       |
| $x^6 - 3x^3 - 3x - 3$              | $6T_{16}$ | $S_6$                    | 720   | no       |

These agree with the GAP enumeration of Tables 1 and 2.

**A.2. Raw GAP — only for the transitive-group library.** The polynomial-side Galois groups were computed in native Sage; GAP was used only where the transitive-group library is authoritative:

```
LoadPackage( "TransGrp" );           # or the transitive_groups package
g := TransitiveGroup( 6, 12 );       # the transitive A5 (6T12)
Size( g );                          # -> 60
IsSolvableGroup( g );               # -> false
StructureDescription( g );          # -> "A5"
Stabilizer( g, 1 );                 # point stabilizer, e.g. D10
```

The full enumeration of the five and sixteen groups, and the cycle-type sets used in §6.1, are in the scripts `enumerate2.g`, `cyc5.g`, and `cyc6_all.g` in the accompanying log.

**A.3. Why the split.** The split between "native Sage for polynomials" and "raw GAP for the group library" is deliberate. Sage's `f.galois_group()` handles the polynomial-side reduction, resolvent, and group-recognition steps end-to-end and is the right tool for the examples of §5. GAP is the authoritative source for the *list* of transitive groups and their representations (point stabilizers, cycle-type sets), which is what Tables 1 and 2 and §6.1 depend on. Driving GAP through Sage's `gap()` interface is possible but fragile (string marshalling and version drift), so the GAP scripts were run standalone against the `gapsystem/gap-docker` image.

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